

BioSpecInfo

Software Architecture Document (SAD) — BioSpecInfo

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Versione **bsi-v187** · commit `c2e30fa`

Generato il 2026-10-07

Documento generato automaticamente dal sorgente Markdown versionato nel repository. La versione e il commit stampati in ogni piè di pagina identificano esattamente lo stato del software descritto.

Software Architecture Document (SAD) — BioSpecInfo

Field	Value
Project	BioSpecInfo
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Type	Progressive Web App (PWA), client-side, local-first, offline-first
Platform	Modern browsers (Chromium, Firefox, Safari/WebKit) · Desktop and Mobile
Repository	<code>samupropiol-ship-it/BioSpecInfo-v11</code>

1. System overview

BioSpecInfo is a **monolithic yet modular** web platform that runs all of its application logic inside the client browser, with no application server. It is distributed as an installable **Progressive Web App** (`display: standalone`) and works **offline** thanks to a Service Worker that precaches the heavy assets.

Architectural pattern: Single-Page/Single-File Application + satellite sub-apps in iframes, with a WebAssembly-based cheminformatics *core* decoupled from the presentation logic.

1.1 Guiding principles

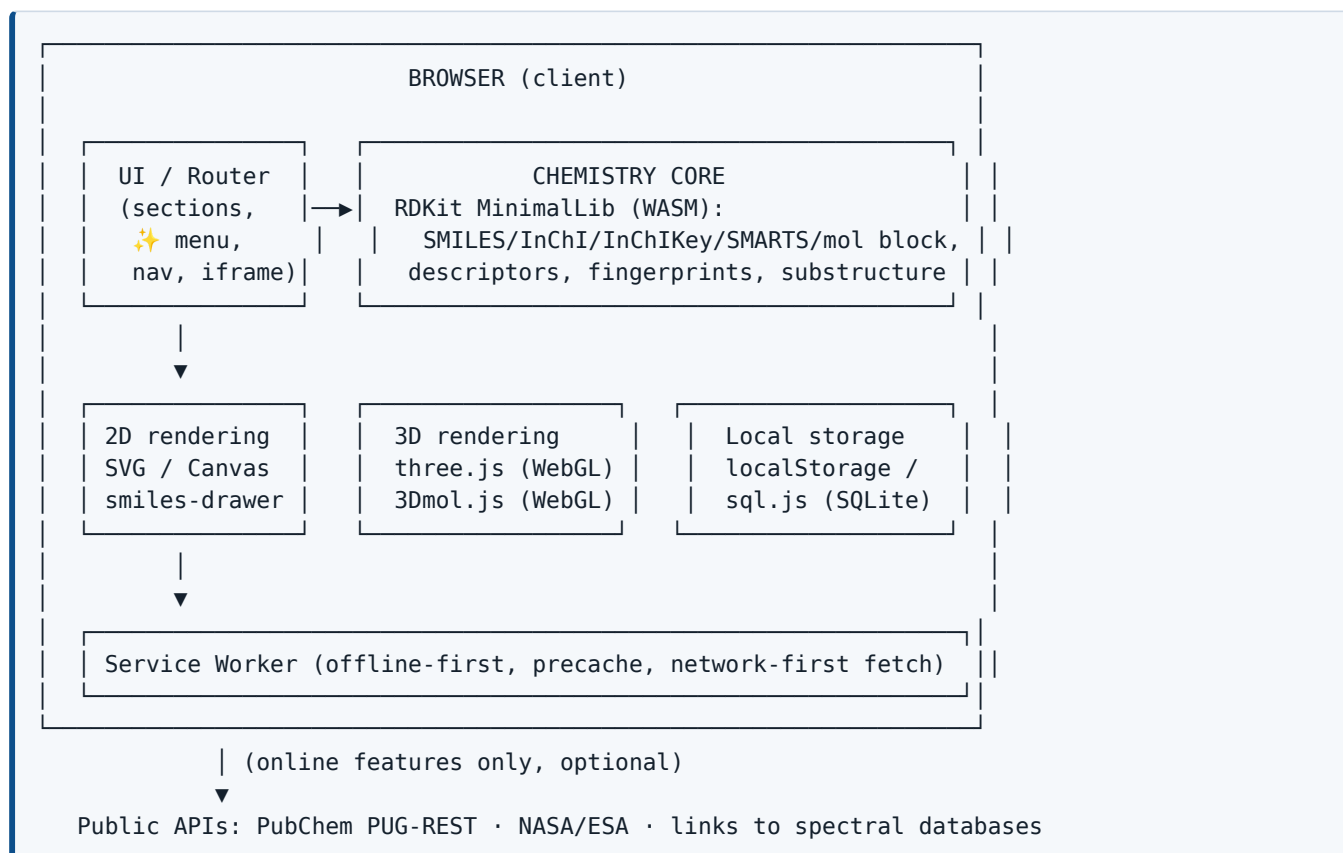
- **Local-First:** data and computations live on the device; no mandatory transmission to third parties.
- **Zero-backend:** no server database, no dynamic hosting cost (deployable on static hosting, e.g. GitHub Pages).
- **Progressive enhancement:** online features (PubChem search, real 3D models, astronomical imagery) degrade gracefully when offline.

2. File structure

File	Role
<code>index.html</code>	Main application (chemistry core, Spectra Centre, Data Science, ✨ menu). ~4.5 MB, over 43,000 lines.
<code>chimorga.html</code>	Organic Chemistry course (25 modules).
<code>astro.html</code>	3D Astrochemistry (three.js).

File	Role
pro.html, rdkit_lab.html, accademia.html, simulazioni.html, guidaret.html, file_manager.html, Biochimica_Guida_Definitiva.html, sr_completo.html, sr_essenziale.html, download.html, changelog_tesi.html	Thematic sub-apps loaded in iframes with ?v=timestamp cache-busting.
manifest.json	PWA configuration (icons, theme, standalone).
sw.js	Service Worker: <i>network-first</i> strategy with cache fallback; precache of heavy libraries/assets; cache version bumped on every release.
RDKit_minimal.js / RDKit_minimal.wasm	Cheminformatics engine (RDKit MinimalLib).
3Dmol-min.js	3D viewer for molecules/macromolecules.
three.min.js, three_bloom.js, gltf_loader.js	WebGL 3D rendering (anatomy, astrochemistry, mechanisms).
smiles-drawer.min.js	2D depiction of SMILES.
lib/sql-wasm.js / lib/sql-wasm.wasm	SQLite in WebAssembly (SQL playground).
models/, textures/	3D assets (organs, écorché, planets) and textures.

3. Component architecture



3.1 Module Router / UI

Navigation between sections via CSS classes (`.section.on`) and a ✨ menu (FAB) that opens the special tools (Spectra Centre, Retrosynthesis, Quantum Lab, Data Science, etc.). Sub-apps are isolated in versioned `iframes`.

3.2 WebAssembly Engine (RDKit MinimalLib)

Loaded on demand (`window.bsiLoadRDKit`) with `locateFile` pointing to the local `.wasm` files and a CDN as a last-resort fallback. It exposes:

- **Identifiers:** canonical/isomeric SMILES, CXSMILES, InChI, InChIKey, SMARTS, MDL V2000 mol block.

- **Descriptors:** molecular weight, monoisotopic exact mass, LogP (Wildman-Crippen), TPSA, HBD/HBA, rotatable bonds, aromatic/aliphatic rings, Fsp³, molar refractivity, stereocentres.

- **Fingerprints & similarity:** Morgan ECFP4 (1024 bit) and the Tanimoto index.
- **Substructure matching** via SMARTS (`get_qmol` + `get_substruct_matches`).
- **2D depiction** (`get_svg` , `get_svg_with_highlights`) and 2D coordinate generation (mol block) re-importable into the editor.

3.3 Rendering

- **2D:** proprietary SVG (reaction mechanisms with Bézier arrows), Canvas 2D (molecule editor, CIP stereochemistry viewer), smiles-drawer.
- **3D:** `three.js` (WebGL) for anatomy (Drug Atlas), astrochemistry and 3D mechanisms; `3Dmol.js` for molecular models from PubChem SDF.

3.4 Storage Layer

- `localStorage` (with a **safety guard** and in-memory fallback for incognito / blocked-storage scenarios).
- `sql.js` (SQLite compiled to WASM) for the Data Science SQL playground.

4. Implemented features (current state)

4.1 Spectra Centre (molecular editor & analysis)

Draw from scratch (full periodic table, ring/group templates, drag-to-bond with snapping) or load via SMILES / library (63 molecules) → full analysis:

- **All chemical notations** (formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS) with one-click copy.

- **Predicted spectra** as sub-panels: IR, ^1H -NMR, ^{13}C -NMR, MS with ****isotope pattern (IUPAC abundances) and adduct masses**** ($[\text{M}+\text{H}]^+$, $[\text{M}+\text{Na}]^+$...), UV-Vis, Raman — each with a bibliographic source.
- **Functional groups** (~30, via SMARTS) and **properties** (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.
- **Drug-likeness / ADMET**: Lipinski, Veber, Egan, Ghose, Muegge rules; estimated ESOL solubility; QED (when exposed by the engine); **Brenk/PAINS structural alerts**; physicochemical radar profile.
- **Elemental analysis** (weight %) and **Tanimoto similarity** against the library.
- **3D models** (ball & stick, spacefill, wireframe) from PubChem; SVG/PNG/MOL export.

4.2 Synthesis & Retrosynthesis

46 disconnection strategies with an **animated mechanism player** in **2D and 3D**: curved *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

4.3 Organic Chemistry (`chimorga.html`)

25 modules: fundamentals, stereochemistry, alkenes/alkynes, aromatics & EAS, SN/E, carbonyls, amines, carbohydrates, amino acids, lipids, nucleotides, animated syntheses, named reactions, heterocycles, pericyclic & Woodward-Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups.

4.4 Data Science

3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), interactive ML labs, SQL playground (`sql.js`), course roadmap.

4.5 Other sections

3D astrochemistry, Quantum Lab, Academy (gamified quizzes with SM-2 spaced repetition), interactive simulations, guides, databases of elements/drugs/ biomolecules.

5. Architectural roadmap (NOT yet implemented)

The following items belong to the *enterprise/HPC* vision and require external binaries, models or build pipelines that cannot be hosted inside a single-file PWA:

- HPC parallelism via Web Workers + `SharedArrayBuffer` / `Atomics` .
- WebGPU acceleration for dense/sparse algebra and GNN inference.
- ML predictors in ONNX Runtime WebGPU (GNN for NMR such as ShiftML2/DeepNMR; hERG, BBB, CYP450, LogS).

- In-browser molecular docking (AutoDock Vina/Webina, PDBQT format).
- Semi-empirical quantum chemistry (xTB/GFN2-xTB in WASM).
- Rule-based computational retrosynthesis engine (SMIRKS).
- SQLite-WASM database at the 1,000,000-structure scale and SDF batch processing

via Web Workers.

- Immersive WebXR (VR/AR) visualisation.
 - Native packaging (Android TWA, iOS, Electron/Tauri) and CI/CD pipelines.
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6. Requirements and constraints

- **Browser:** WebAssembly and WebGL support; JavaScript enabled.
- **Network:** not required for the core; recommended for PubChem search and real 3D models.
- **Performance:** target 30–60 fps for interactive rendering; heavy computations are delegated to RDKit WASM and kept responsive.